

Recurrent Neural Networks and LSTMs: Learning from Sequential Data

Modeling sequences, memory, and temporal dependencies in neural networks

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Table of Contents – Motivation

1. Motivation
2. RNNs: Recurrent Neural Networks
 - 2.1 General Introduction
 - 2.2 Vanilla RNNs
 - 2.3 Training of RNNs
 - 2.4 Implementing simple RNNs (Vanilla RNNs) in Tensorflow
3. LSTM: Long Short-Term Memory
 - 3.1 Motivation
 - 3.2 LSTM Architecture
 - 3.3 Implementing LSTMs in Tensorflow

Motivation

Motivation

- **Not all problems** can be converted into one with **fixed-length inputs and outputs**.
- Some tasks, such as **Speech Recognition** or **Time-series Prediction**, require a system to **store and use context information**.
- **Example:** Output **YES** if the number of 1s is even, else **NO**:
 - 1000010101 → **YES**
 - 100011 → **NO**
- It is **hard or impossible** to choose a fixed context window:
 - There can always be a **new sample longer** than anything seen before.

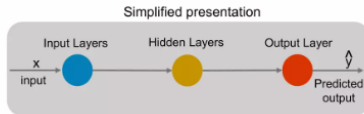
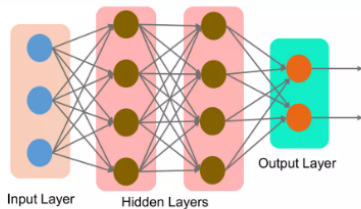
Recap: Feedforward Neural Networks (FNNs)

Generally, when we talk about **simple neural networks**, they typically mean **feedforward neural networks (FNNs)** — especially the multi-layer perceptron (MLP) type. These are the foundational architectures used in many early machine learning models.

Recap: Feedforward Neural Networks (FNNs)

Generally, when we talk about **simple neural networks**, they typically mean **feedforward neural networks (FNNs)** — especially the multi-layer perceptron (MLP) type. These are the foundational architectures used in many early machine learning models.

In a Feed-Forward Network, information flows only in forward direction, from the input nodes, through the hidden layers (if any) and to the output nodes. There are no cycles or loops in the network.



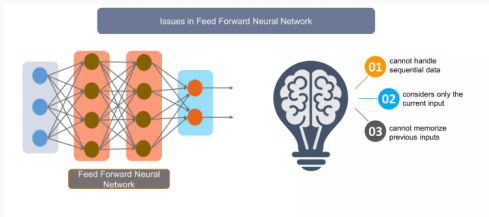
- Decisions are based on current input
- No memory about the past
- No future scope

Limitations of FNNs

- FNNs treat each input as independent — they lack any notion of order or context.
- No mechanism to retain information about previous inputs or time steps.
- Fails on tasks where input sequence matters, such as:
 - Natural Language: “The cat sat” $\neq$ “Sat the cat”
 - Time-Series: Predictions require trends from earlier time points
 - Speech/Audio: Context and timing are crucial

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FNNs process each input independently: context is lost.

Motivation for RNNs and LSTMs

We need models that can:

- Learn from sequences — where order and context matter
- Retain and reuse past information — like memory
- Adjust their understanding as more input is seen

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- Introduce a feedback loop — current output depends on past hidden states
- Suitable for sequence modeling tasks (language, time-series)
- Struggle with **long-term dependencies** due to vanishing gradients

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Long Short-Term Memory (LSTM):

- Enhances RNNs with **memory cells** and **gating mechanisms**
- Capable of learning **long-range dependencies** effectively
- Avoids vanishing gradient issues via internal memory regulation

RNNs: Recurrent Neural Networks

2. RNNs: Recurrent Neural Networks

2.1 General Introduction

2.2 Vanilla RNNs

2.3 Training of RNNs

2.4 Implementing simple RNNs (Vanilla RNNs) in Tensorflow

What is a Recurrent Neural Networks (RNNs)?

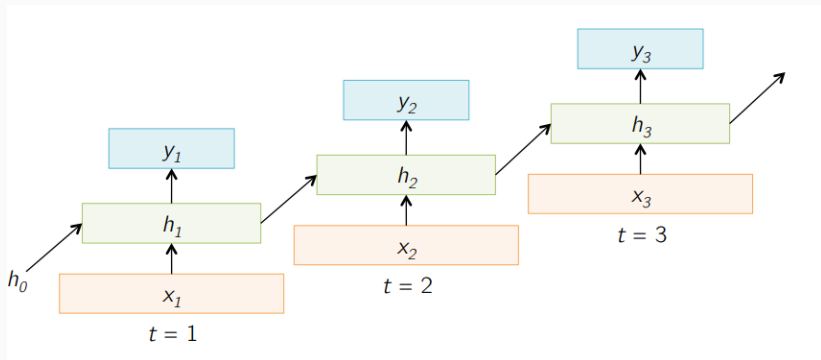
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What is a Recurrent Neural Networks (RNNs)?

RNNs are a class of neural networks designed for modeling **sequential data**. They maintain an internal memory to capture temporal dynamics.

- The output at time t depends on the input at t and the hidden state from $t - 1$. That means: RNNs take the **previous output** or **hidden states** as part of the input at each time step.
- The composite input at time t carries **historical information** from earlier time steps ($T < t$).
- RNNs are powerful because their intermediate values (also called **hidden states**) can **store information** about past inputs for a time that is **not fixed a priori**.
- Ideal for problems where **context or order matters**
- Used in language modeling, speech, and time-series tasks

Sample RNNs



Where

- h_t is the hidden state at time t
- x_t is the input vector at time t

2. RNNs: Recurrent Neural Networks

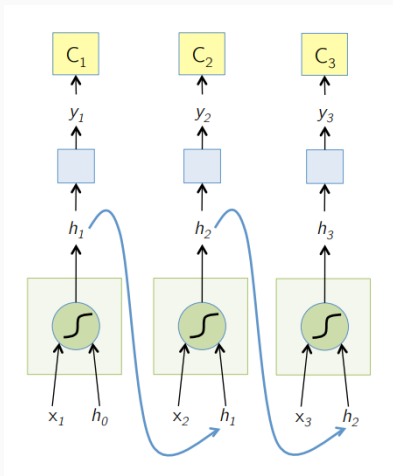
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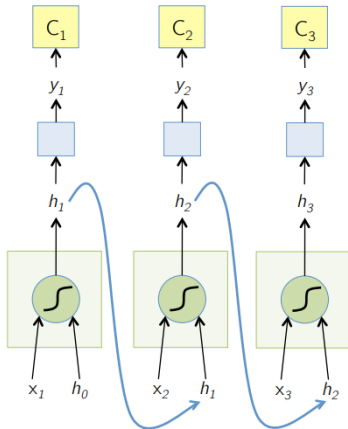
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RNNs Forward Steps for (for Vanilla RNNs)



An RNN processes input sequences one step at a time. The same RNN cell (with shared weights) is reused across all time steps. This makes the RNN look like a deep network with depth equal to the sequence length.

RNNs Forward Steps for (for Vanilla RNNs)



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- The basic RNN cell updates as:

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b)$$

where W_h , W_x , and b are learnable parameters.

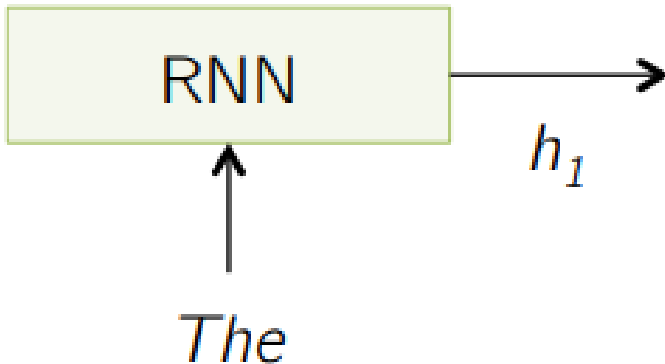
- The output $y_t = \sigma(W_y h_t)$, where $\sigma(\cdot)$ is the activation function for the output layer
- $C_t = \text{Loss}(y_t, \text{GT}_t)$ is the loss calculated at time t
- Note that the weights W_h , W_x , W_y and biases b are shared over time

RNNs application: Sentiment Classification

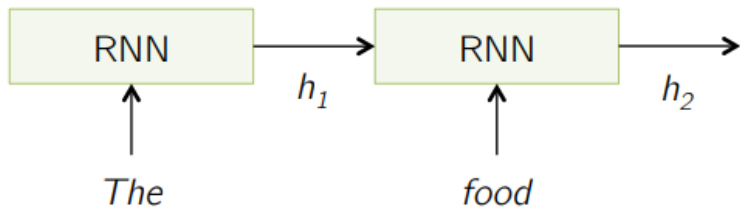
- **Goal:** Classify a review (e.g., from [Yelp!](#), [IMDB](#), etc.) as **positive** or **negative**.
- **Inputs:** Multiple words forming one or more sentences.
- **Outputs:** Binary classification — Positive / Negative.
- **Examples:**
 - *"The food was really good."* ⇒ **Positive**
 - *"The chicken crossed the road because it was uncooked."* ⇒ **Negative**
 - *"Absolutely loved the cinematography and acting."* ⇒ **Positive**
 - *"The plot was dull and completely predictable."* ⇒ **Negative**
 - *"Would definitely come here again!"* ⇒ **Positive**
 - *"Terrible service and cold food."* ⇒ **Negative**

Sentiment Classification I

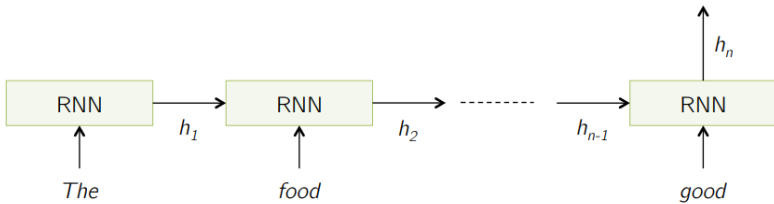
Consider the input: "The food was really good"



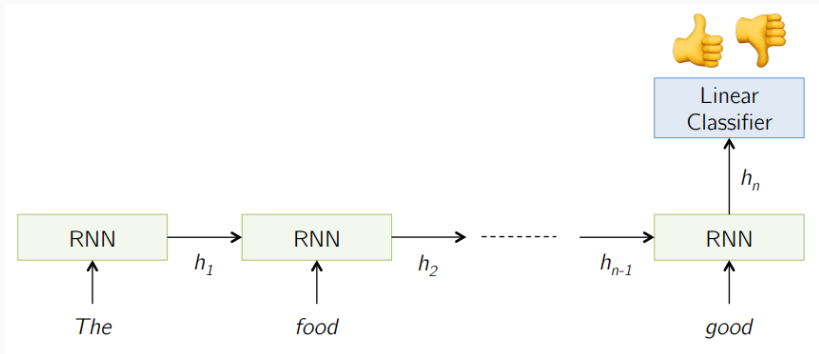
Sentiment Classification II



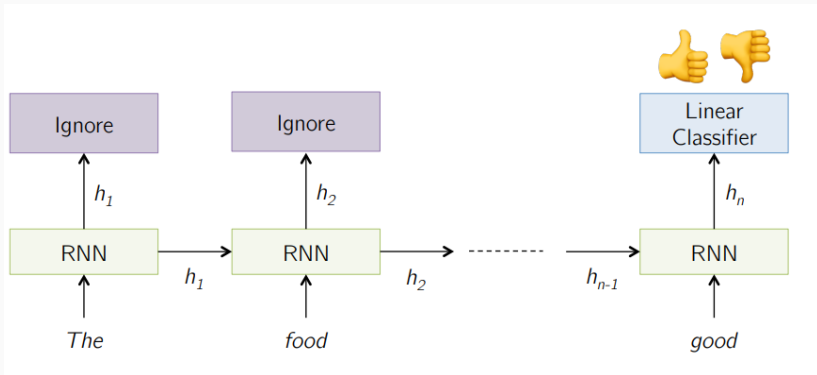
Sentiment Classification III



Sentiment Classification IV



Sentiment Classification V



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RNNs Backward Steps: Backpropagation Through Time (BPTT)

- Recall the **forward RNN** steps:

- Calculation at t -th hidden state

$$h_t = \tanh(W_h h_{t-1} + W_x x_t + b) \quad \text{and} \quad y_t = \sigma(W_y h_t)$$

- The loss at time t :

$$C_t = \text{Loss}(y_t, \text{GT}_t)$$

- Total loss across T time steps:

$$C = \sum_{t=1}^T C_t$$

- The Backpropagation Step:**

- Compute gradients $\frac{\partial C}{\partial W_h}, \frac{\partial C}{\partial W_x}, \frac{\partial C}{\partial W_y}, \frac{\partial C}{\partial b}$
- Requires unrolling the network over time and applying the chain rule recursively

Gradient Flow Through Time

- Gradients are propagated backward from time T to $t = 1$.
- Chain rule applied recursively:

$$\frac{\partial C}{\partial W_h} = \sum_{t=1}^T \frac{\partial C_t}{\partial h_t} \cdot \frac{\partial h_t}{\partial W_h}$$

- But $\frac{\partial h_t}{\partial W_h}$ depends on h_{t-1} , which depends on h_{t-2} , and so on.
- So gradients accumulate via:

$$\frac{\partial C_t}{\partial h_k} = \frac{\partial C_t}{\partial h_t} \cdot \prod_{j=k+1}^t \frac{\partial h_j}{\partial h_{j-1}}, \quad \text{for } k < t$$

and leads to numerical instability during training

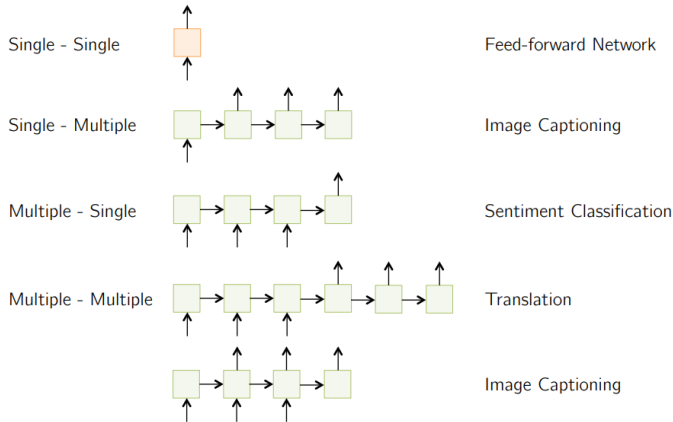
Mitigating Exploding Gradients

- **Gradient Clipping:**

$$\text{If } \|\nabla C\| > \tau, \quad \nabla C \leftarrow \tau \cdot \frac{\nabla C}{\|\nabla C\|}$$

- **Layer Norm / Batch Norm:** Normalize intermediate states
- **Architecture Changes:**
 - Use LSTMs or GRUs — designed to handle long-term dependencies better
 - Residual (skip) connections (in some variants)

Input-Output Scenarios in RNNs



We might deliberately choose to frame our problem as a particular input-output scenario for ease of training or better performance.

Limitations of RNNs

- **Vanishing gradients** — difficult to learn long-range dependencies
- **Exploding gradients** — unstable training
- Difficulty retaining information across long sequences
- Struggles with tasks requiring memory over many steps

These limitations lead to poor performance on tasks like language modeling and sequence prediction.

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Implementing RNNs in TensorFlow

We demonstrate how to build a simple RNN using `tf.keras` to perform time series prediction. Open the Google Colab notebook from the Link:

https://github.com/yadavrishikesh/Masterclass-IITMandi/blob/main/Module4/code/NNs_introduction_to_RNN_GColab.ipynb

Step 1: Prepare the Data

- Use a univariate or multivariate time series
- Convert it to input/output sequences using sliding windows

```
1 import numpy as np
2
3 # Example: univariate time series
4 series = np.sin(np.linspace(0, 100, 1000))
5
6 # Prepare X, y sequences
7 def create_sequences(series, window):
8     X, y = [], []
9     for i in range(len(series) - window):
10         X.append(series[i:i+window])
11         y.append(series[i+window])
12     return np.array(X), np.array(y)
13
14 window = 20
15 X, y = create_sequences(series, window)
16 X = X[..., np.newaxis] # shape: (samples, timesteps,
    features=1)
```

Step 2: Build the RNN Model

- Use SimpleRNN for sequential modeling
- Output a single value for each sequence

```
1 model = tf.keras.Sequential([  
2     tf.keras.layers.SimpleRNN(50, activation='tanh',  
3         input_shape=(X.shape[1], X.shape[2])),  
4     tf.keras.layers.Dense(1)  
5 ])   
6 model.compile(optimizer='adam', loss='mse')  
7 model.summary()
```

Step 3: Train the RNN Model

- Fit the model to predict future values from past sequences
- Use a validation set to monitor performance

```
1 from sklearn.model_selection import train_test_split
2
3 X_train, X_val, y_train, y_val = train_test_split(
4     X, y, test_size=0.2, shuffle=False)
5
6 model.fit(X_train, y_train,
7           validation_data=(X_val, y_val),
8           epochs=20, batch_size=32)
```

Step 4: Make Predictions

- Predict the next value(s) in the sequence

```
1 y_pred = model.predict(X_val)
2
3 # Plot predictions
4 import matplotlib.pyplot as plt
5
6 plt.plot(y_val[:100], label='True')
7 plt.plot(y_pred[:100], label='Predicted')
8 plt.legend(); plt.title("Time Series Prediction")
```

Customization Notes and Summary

- **Customization Notes**

- **Initial states:** Use `initial_state=` argument in `SimpleRNN()` if needed
- **Stacked RNNs:** Set `return_sequences=True` in lower layers
- **Dropout:** `dropout=` and `recurrent_dropout=` to regularize
- **Cell-level API:** Use `tf.keras.layers.RNN(cell)` for full customization
- **Gradients:** TensorFlow automatically handles BPTT

- **Summary**

- RNNs are useful for sequence modeling tasks
- TensorFlow provides high-level RNN APIs (`SimpleRNN`, `LSTM`, `GRU`)
- Training requires careful preprocessing and attention to input shape
- For long sequences, consider `LSTMs` or `GRUs` for better stability

LSTM: Long Short-Term Memory

3. LSTM: Long Short-Term Memory

3.1 Motivation

3.2 LSTM Architecture

3.3 Implementing LSTMs in Tensorflow

Why Do We Need LSTM?

Standard RNNs suffer from:

- **Vanishing/exploding gradients**: hard to learn long dependencies
- **Short-term memory**: forgets earlier inputs quickly

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LSTM can preserve information over **long sequences** and

- Decide what to **remember**
- Decide what to **forget**
- Decide what to **output**

LSTM: Long Short-Term Memory

LSTM is a special type of Recurrent Neural Network (RNN) designed to learn long-term dependencies. It addresses the limitations of standard RNNs which struggle with vanishing and exploding gradients.

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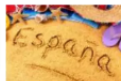
- Introduced by Hochreiter and Schmidhuber (1997)
- Effective at learning long-range sequences
- Widely used in NLP, speech recognition, and time series forecasting

Why Long-Term Dependencies is Important?

Suppose we try to predict the last word in the text



"I have been staying in Spain for the last 10 years... I can speak fluent Spanish"



The word we predict will depend on the previous few words in context

- Here we need the context of **Spain** to predict the last word in the text.
- It's possible that the gap between the relevant information and the point where it is needed to become very large.
- **LSTMs** help us solve this problem.

3. LSTM: Long Short-Term Memory

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LSTM Architecture (1)

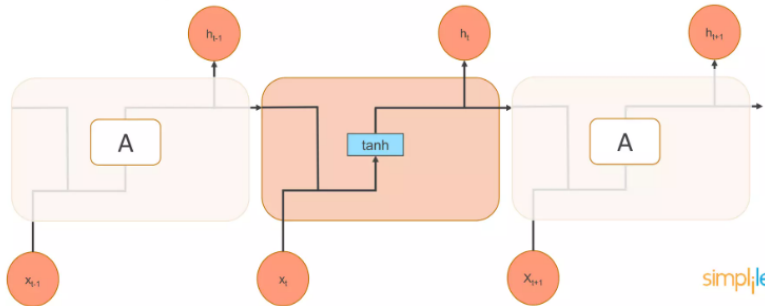
- The key feature is the **cell state** (C_t), which runs through the entire chain with only minor interactions.
- Information is added or removed via **gates**:
 - **Forget gate** f_t
 - **Input gate** i_t
 - **Output gate** o_t

Hidden state h_t carries the output at each time step.

LSTM Architecture (2)

LSTMs are special kind of Recurrent Neural Networks, capable of learning long-term dependencies. Remembering information for long periods of time is their default behavior.

All recurrent neural networks have the form of a chain of repeating modules of neural network. In standard RNNs, this repeating module will have a very simple structure, such as a single tanh layer.



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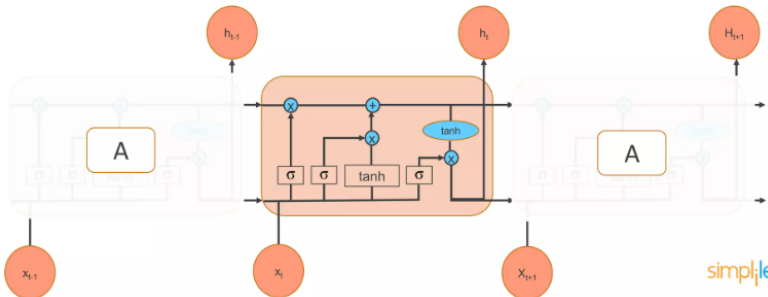
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¹<https://www.simplilearn.com/>

LSTM Architecture (3)

LSTMs are special kind of Recurrent Neural Networks, capable of learning long-term dependencies. Remembering information for long periods of time is their default behavior.

LSTMs also have a chain like structure, but the repeating module has a different structure. Instead of having a single neural network layer, there are four interacting layers communicating in a very special way.

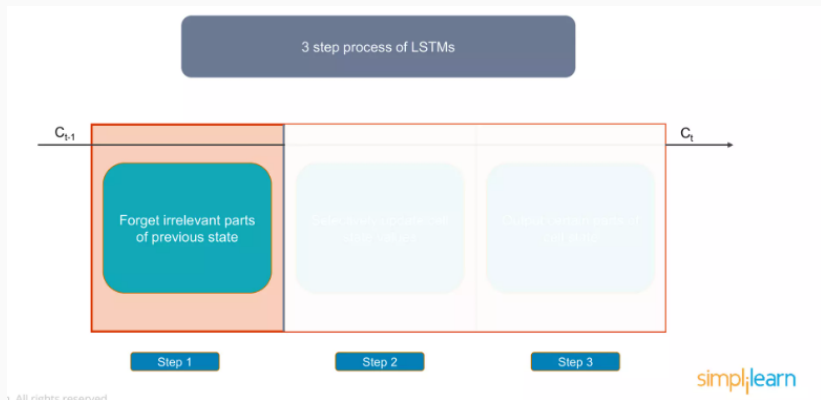


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²<https://www.simplilearn.com/>

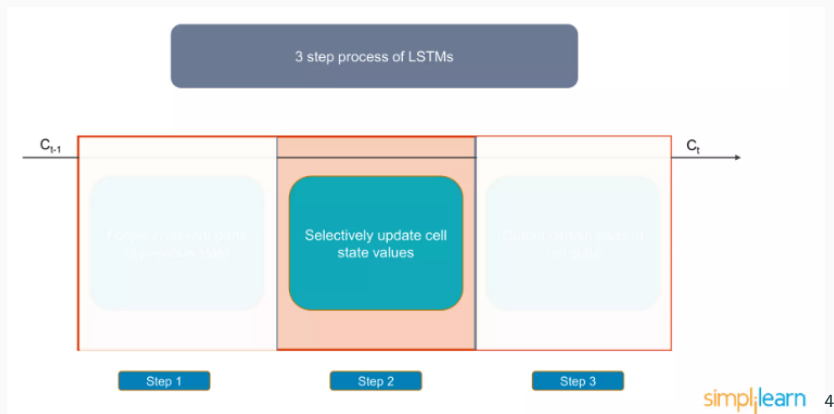
LSTM Architecture (4)



1. All rights reserved.

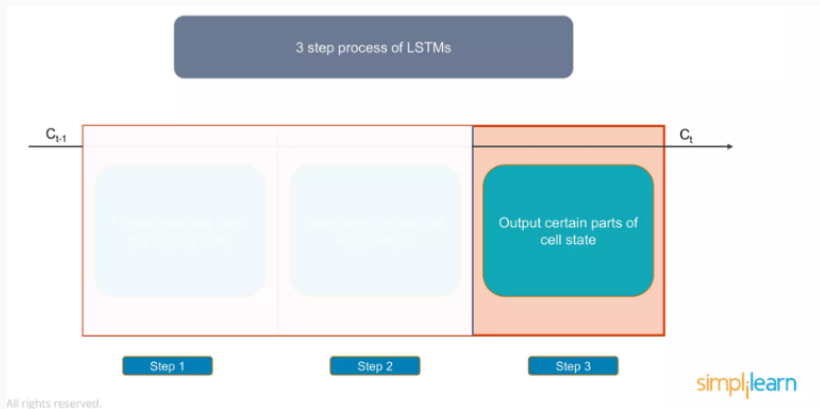
³<https://www.simplilearn.com/>

LSTM Architecture (5)



⁴<https://www.simplilearn.com/>

LSTM Architecture (6)



⁵<https://www.simplilearn.com/>

Working of LSTMs Cell (1)

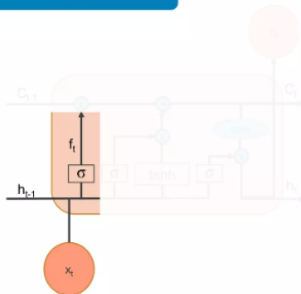
Step-1

Decides how much of the past it should remember

First step in the LSTM is to decide which information to be omitted in from the cell in that particular time step. It is decided by the sigmoid function. It looks at the previous state (h_{t-1}) and the current input x_t and computes the function.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

f_t = forget gate
Decides which information to delete
that is not important from previous
time step



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⁶<https://www.simplilearn.com/>

Working of LSTMs Cell (2)

Step-2

Decides how much should this unit add to the current state

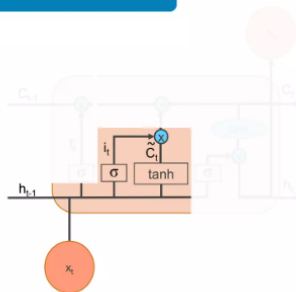
In the second layer, there are 2 parts. One is the sigmoid function and the other is the tanh. In the **sigmoid** function, it decides which values to let through (0 or 1). **tanh** function gives the weightage to the values which are passed deciding their level of importance (-1 to 1).

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C)$$

i_t = input gate

Determines which information to let through based on its significance in the current time step



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⁷<https://www.simplilearn.com/>

Working of LSTMs Cell (3)

Step-3

Decides what part of the current cell state makes it to the output

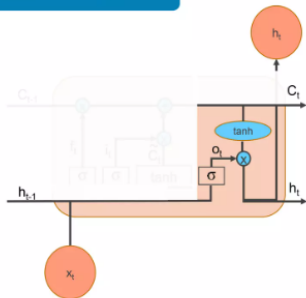
The third step is to decide what will be our output. First, we run a sigmoid layer which decides what parts of the cell state make it to the output. Then, we put the cell state through tanh to push the values to be between -1 and 1 and multiply it by the output of the sigmoid gate.

$$o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t * \tanh(C_t)$$

o_t = output gate

Allows the passed in information to impact the output in the current time step



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Working of LSTMs Cell (4): LSTM Cell Equations

At time step t , given input x_t , previous hidden state h_{t-1} , and previous cell state C_{t-1} :

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (\text{Forget Gate})$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (\text{Input Gate})$$

$$\tilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \quad (\text{Candidate Cell State})$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (\text{Updated Cell State})$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (\text{Output Gate})$$

$$h_t = o_t \odot \tanh(C_t) \quad (\text{Hidden State Output})$$

σ is the sigmoid activation; $\odot$ is element-wise multiplication.

How LSTM Is Trained

- **Forward Pass:**

- Input sequences are passed through the LSTM layer(s).
- At each time step, hidden and cell states are updated using gates.
- The final output is passed to a dense layer to compute predictions.
- A loss (e.g., MSE or cross-entropy) is calculated from predictions.

- **Backward Pass:**

- Gradients of the loss are computed using *Backpropagation Through Time (BPTT)*.
- Gradients flow backward across all time steps and LSTM parameters.
- Optimizer (e.g., Adam) updates weights to minimize the loss.

- **Repeat for multiple epochs** until validation loss stabilizes or early stopping is triggered.

Advantages of LSTM

- **Handles long sequences:** Able to learn long-term dependencies
- **Gradient flow:** Mitigates vanishing gradient problem
- **Memory mechanism:** Gating allows selective memory and forgetting
- **Versatile:** Works well with sequences of varying length

Compared to traditional RNNs, **LSTMs** are much more stable and effective during training.

LSTM networks are widely used in sequential data problems:

- Natural Language Processing (NLP):
 - Text generation
 - Machine translation
 - Named entity recognition
- Speech Recognition
- Time Series Forecasting
- Video Analysis

LSTM remains a foundational component in many deep learning applications involving temporal data.

3. LSTM: Long Short-Term Memory

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Implementing LSTMs in TensorFlow

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Preparing Time Series Data for LSTM

- LSTM expects input as 3-D tensors: (samples, timesteps, features).
- Convert raw time series into sequences using a sliding window.

```
1 import numpy as np
2
3 def create_sequences(data, window_size):
4     X, y = [], []
5     for i in range(len(data) - window_size):
6         X.append(data[i:i+window_size])
7         y.append(data[i+window_size])
8     return np.array(X), np.array(y)
9
10 X, y = create_sequences(my_series, window_size=10)
11 X = X[..., np.newaxis]          # Add feature dim for
    univariate
```

LSTM Model Architecture

```
1 from tensorflow.keras.models import Sequential
2 from tensorflow.keras.layers import LSTM, Dense
3
4 model = Sequential([
5     LSTM(64, activation='tanh',
6         input_shape=(X.shape[1], X.shape[2])),
7     Dense(1) # Predict next value
8 ])
9
10 model.compile(optimizer='adam', loss='mse')
11 model.summary()
```

- Stack additional LSTMs or add dropout if needed.
- Adjust the output layer for regression or classification tasks.

Model Training and Evaluation

```
1 history = model.fit(  
2     X_train, y_train,  
3     validation_data=(X_val, y_val),  
4     epochs=20,  
5     batch_size=32,  
6     verbose=1  
7 )  
8  
9 loss = model.evaluate(X_test, y_test)  
10 print(f"Test loss: {loss:.3f}")
```

- Monitor validation loss to avoid over-fitting.
- Consider callbacks like EarlyStopping or ReduceLROnPlateau.

Making Forecasts

```
1 last_sequence = X_test[-1].reshape(1, X.shape[1], X.shape  
   [2])  
2 predicted = model.predict(last_sequence)  
3 print(f"Predicted next value: {predicted[0,0]:.2f}")
```

- Roll forward one step at a time for multi-step forecasts.
- Feed predictions back as input for iterative forecasting.